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A novel dynamic signal Lemma for predefined-time stabilization of high-order nonlinear systems with dynamic uncertainties

Jiawei Ma ^a, Huaguang Zhang ^{a,b,*}, Juan Zhang ^a, Le Wang ^a

^a College of Information Science and Engineering, Northeastern University, China

^b State Key Laboratory of Synthetical Automation for Process Industries (Northeastern University), China

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ABSTRACT

This research considers the problem of predefined-time adaptive neural control for high-order nonlinear systems with dynamics uncertainties. Through the application of neural networks, the unknown nonlinear functions of the systems are approximated. In order to avoid the “complexity explosion problem”, a novel predefined-time dynamics surface is designed for high-order nonlinear systems. Regarding the dynamics uncertainties existing within the systems, this paper designs and introduces a modified predefined-time dynamics signal to address the aforementioned issue. By using a backstepping design scheme, the neural predefined-time control approach is proposed so that all signals in considered systems are bounded within a predefined time. To verify the validity of the proposed method, an illustrative example is presented.

1. Introduction

The high-order nonlinear systems (HNSs) represent an advancement and expansion of strict-feedback nonlinear systems. This class is characterized by higher-order powers, distinguishing it from typical nonlinear systems. In practical scenarios, high-order nonlinear systems can be described by the following practical systems, such as: the under-actuated, weakly coupled and unstable mechanical system (Liu et al., 2015), and the second-order dynamical model of a boiler-turbine unit (Rui et al., 1997). However, powers greater than one in high-order nonlinear systems lead to uncontrollable Jacobian linearization, rendering feedback linearization infeasible (Wu et al., 2022). Hence, control of high-order nonlinear systems remains challenging, but the adding power integral technique offers an effective solution (Lin & Qian, 2000).

Due to their inherent approximation capabilities, neural logic systems (FLSs) and neural networks (NNs) are frequently employed as common approximators to approximate unknown nonlinear functions within systems. Through their support, several achievements have been documented (Ouyang et al., 2025; Zhang et al., 2023a, 2010, 2023c). As is widely known, the iterative differentiation of the virtual controller in the process of applying the backstepping method for controller design introduces intricate and laborious computations, leading to the emergence of the “complexity explosion problem”. It is fortunate that the advent of dynamics surface techniques have provided a solution to the quandary (Hu & Ma, 2025; Kwan & Lewis, 2000; Peng & Dubay, 2019; Tu et al., 2026; Yip & Hedrick, 1998).

The settling time represents a crucial aspect to consider when investigating control problems. The limited and controllable settling time is often a preferred option for users. Bhat et al. first introduced the concept of finite-time control (Bhat & Bernstein, 2000), which subsequently spurred significant developments (Zhang et al., 2021a,b). Although finite-time control exhibits characteristics such as faster convergence speed and stronger anti-interference capability, its settling time is strongly dependent on the initial state of the system, and the upper bound of the settling time varies with the initial state and cannot be arbitrarily specified. Following this, the introduction of fixed-time control broke through this impasse. Fixed-time control is a further extension of finite-time control, whose settling time is independent of initial conditions and determined solely by the fixed design parameters in the controller. There have been noteworthy achievements in fixed-time control in recent years (Li et al., 2026; Ma et al., 2024; Zhang et al., 2025). Although the settling time of fixed-time control is no longer dependent on the system's initial conditions, the complexity arises from the multitude of design parameters upon which fixed-time control relies, and the intricate relationships among these parameters remain elusive. Moreover, since the upper bound of the settling time is dictated by the inherent design parameters, it still cannot be directly specified. Consequently, a more refined concept, designated as predefined-time control, has been developed. In this method, the settling time depends only on a single value directly specified in advance by the user, neither on the initial state of the system nor on the complex parameters of the system, and the upper bound of the settling time can be freely and flexibly set by the

* Corresponding author.

E-mail addresses: jiaweima1998@163.com (J. Ma), hgzhang@ieee.org (H. Zhang), zjneu11@163.com (J. Zhang), wangle9909@163.com (L. Wang).

designer (Ma et al., 2025; Xie & Chen, 2022; Yang et al., 2025). It should be noted that the aforementioned results do not address the unmodeled dynamics and dynamic disturbances.

In practical industrial systems, due to the widespread presence of modeling errors, modeling simplifications, and measurement noise, the process of modeling the controlled plant is often accompanied by unmodeled dynamic uncertainties, such as parasitic capacitances and inductances in electrical circuit systems. If not properly addressed, these uncertainties may not only degrade control performance but also lead to instability of the closed-loop system. In response to this issue, numerous scholars have conducted extensive research (Ai et al., 2024; Tong et al., 2021; Wang et al., 2023). In Wang et al. (2023), the finite-time dynamic signal was presented as a means of coping with the dynamic disturbances. The problem of the robust neural adaptive finite-time control design for HNSs with unmodeled dynamics was studied from Tong et al. (2021). Furthermore, the neural adaptive fixed-time control for stochastic nonlinear systems with unmodeled dynamics was considered (Ai et al., 2024). In addition, many control systems are prone to actuator faults during operation, which may lead to system instability. Taking dynamical boiler-turbine units as an example, faults will directly disrupt the energy and power regulation mechanism and threaten the structural safety of the boiler and the turbine. Thus, there have been a significant focus on the development of fault-tolerant control (Wang & Xia, 2024; Zhang et al., 2023b).

To the best of our knowledge, the existing literature exhibits a notable scarcity of studies addressing the problem of predefined-time control for high-order nonlinear systems with dynamics uncertainties and actuator fault. The existing technology is unable to provide a direct approach to overcoming these obstacles, presenting a number of challenges, including the design of a dynamic signal that conforms to the predefined-time theory model in order to address dynamics uncertainties in the system. How to design a dynamic surface filter that meets the criteria of the predefined-time theory model and is applicable to high-order nonlinear systems. In view of these unresolved issues, our research addresses the predefined-time control problem of high-order nonlinear systems with dynamic uncertainties and actuator fault. The introduction of NNs enables the approximation of unknown nonlinear functions within systems. A predefined-time dynamic signal is designed and provided as evidence to address the unknown uncertainties of the system. The predefined-time dynamics surface control technology can be developed to cope with "complexity explosion problem". Then, an adaptive neural predefined-time control scheme is presented for high-order nonlinear systems with dynamics uncertainties and actuator fault. The essential novel aspects of this paper are as follows:

(I) This paper first designs a predefined-time dynamic signal and provides a corresponding lemma to address the challenge of dynamic uncertainties in controller design. The upper bound of the convergence time for this dynamic signal can be directly set by the user according to practical needs, without relying on the system's initial state or other design parameters. Compared with existing methods, one study employs finite-time dynamic signals to handle unmodeled dynamics in systems, where the convergence time depends on the initial state and design parameters (Wang et al., 2023). The other study employs fixed-time dynamic signals (Ai et al., 2024), whose convergence time is independent of the initial state but still constrained by design parameters, with complex inter-parameter relationships. The proposed predefined-time dynamic signal in this paper can not only suppress the impact of dynamic uncertainties on system performance but also converges within a user-specified time, which is solely determined by a single user-defined parameter and can be flexibly adjusted according to requirements.

(II) This paper proposes a predefined-time dynamic surface filter designed to effectively avoid the complexity explosion problem inherent in traditional backstepping methods, while aligning its convergence behavior with the theoretical framework of predefined-time control. Unlike existing dynamic surface filters in the literature, which primarily target finite-time control or general nonlinear systems (Hu & Ma, 2025;

Peng & Dubay, 2019), the proposed predefined-time dynamic surface filter offers two key features. First, its convergence time upper bound can be directly specified by the user, providing flexibility. Second, the filter structure is applicable to high-order nonlinear systems where the powers of the state and control input are positive odd integers, effectively overcoming structural challenges in controller design for such systems. Furthermore, the proposed predefined-time control scheme exhibits generality. Even when the system's exponent term equals 1, meaning the system reduces to a general nonlinear case, the designed controller remains stable and effective.

2. Problem description and preliminaries

2.1. System model

The high-order nonlinear systems is considered as

$$\begin{cases} \dot{z}(t) = q(z, y), \\ \dot{x}_i = x_{i+1}^{p_i} + f_i(\bar{x}_i) + \Delta_i(y, z), \\ \dot{x}_n = u_f^{p_n} + f_n(\bar{x}_n) + \Delta_n(y, z), \\ y = x_1, \end{cases} \quad (1)$$

where $\bar{x}_i = [x_1, \dots, x_i]^T \in \mathbb{R}^i$ ($i = 1, 2, \dots, n-1$) and $\bar{x}_n = [x_1, \dots, x_n]^T \in \mathbb{R}^n$ denote the system states. $p_j \geq 1$ ($j = 1, 2, \dots, n$) are the positive odd integers and the positive odd integer p can be defined as: $p = \max_{j=1, \dots, n} \{p_j\}$.

z is the unmodeled dynamics and Δ_j denote the nonlinear dynamic uncertainties ($j = 1, 2, \dots, n$). Assume that $\Delta_j(\cdot)$ and $q(\cdot)$ are uncertain Lipschitz continuous functions. y denotes output of the high-order systems and $u_f^{p_n}$ is the control input. $f_i(\bar{x}_i)$ and $f_n(\bar{x}_n)$ denote unknown nonlinear functions.

Consider the system (1) with actuator fault: $u_f(t) = \eta_n u + \xi(t)$, where $\xi(t)$ depicts a bounded function, $0 < \eta_n \leq 1$ denotes the remaining control rate and u is the actual controller that needs to be designed. (1) When $\eta_n = 1$, it indicates that the actuator experiences a bias fault. (2) When $\xi(t) = 0$, it indicates that the actuator experiences a gain fault.

Control Target: The control target is to design an adaptive predefined-time neural fault-tolerant controller for the HNSs, such that: (1) the designed controller can sure that the output signal y can follow the trajectory of the leader $y_d(t)$ regardless of whether there is a fault in the actuator; (2) all signals of the high-order nonlinear systems are proven to be predefined-time bounded.

Assumption 1. (Chen et al., 2009): y_d is the given continuous bounded output signal, and $\dot{y}_d$ and $\ddot{y}_d$ are bounded functions.

Assumption 2. (Li et al., 2019a): p_i ($i = 1, 2, \dots, n-1$) are the positive odd integers that can satisfy $\frac{p+1}{p_i} \geq p - p_{i+1} + 1$.

Assumption 3. (Cui & Xiang, 2023): For the dynamic disturbances Δ_i , there exist $\psi_{i1}(\cdot)$ and $\psi_{i2}(\cdot)$, which are known nonnegative smooth functions, such that $|\Delta_i| \leq \psi_{i1}(|y|) + \psi_{i2}(|z|)$ and $i = 1, \dots, n$. Without loss of generality, assume $\psi_{i2}(0) = 0$.

Assumption 4. For the system $\dot{z} = q(z, y)$, there is a Lyapunov function $V(z)$ satisfying:

$$\begin{cases} \gamma_1(|z|) \leq V(z) \leq \gamma_2(|z|), \\ \frac{\partial V(z)}{\partial z} q(z, y) \leq -\frac{\pi}{rT_c} V^{1+\frac{r}{2}}(z) - \frac{\pi}{rT_c} V^{1-\frac{r}{2}}(z) \\ \quad + \gamma(|x_1|) + \bar{d}_o \end{cases} \quad (2)$$

where γ_1, γ_2 and γ are class k_∞ -functions, $0 < r < 1, \pi > 0, T_c > 0, \bar{d}_o > 0$ is a known constant.

Remark 1. It is of significance to highlight that Assumption 4 is the restrictive condition for investigating nonlinear systems with dynamic uncertainties. The implication of Assumption 4 is that z -systems is input-to-state practically stable.

2.2. Predefined-time theory and mathematical Lemmas

Lemma 1. (Xie & Chen, 2022): Consider a continuously differentiable positive definite function $V(x)$: $\dot{V}(x) \leq -\frac{\pi}{rT_c} V^{1-\frac{r}{2}} - \frac{\pi}{rT_c} V^{1+\frac{r}{2}} + d$, where $r \in (0, 1)$, $T_c > 0$, $\pi > 0$ and $d > 0$. The dynamical system $\dot{x} = f(x, t)$ is defined as practical predefined-time stable. The convergence region is $\{\lim_{t \rightarrow T_c} |V| \leq \min\{(\frac{2rT_c d}{\pi})^{\frac{2}{2-r}}, (\frac{2rT_c d}{\pi})^{\frac{2}{2+r}}\}\}$. The settling time $\bar{T}_c < T_q = \sqrt{2}T_c$, where T_q is the upper bound.

Lemma 2. (Qian & Lin, 2001): For $x_1 \in \mathbb{R}, \forall x_2 \in \mathbb{R}, \omega > 0, \rho > 0$ and $\iota > 0$, then $|x_1|^\omega |x_2|^\rho \leq \frac{\omega}{\omega+\rho} \iota |x_1|^{\omega+\rho} + \frac{\rho}{\omega+\rho} \iota^{-\frac{\omega}{\rho}} |x_2|^{\omega+\rho}$.

Lemma 3. (Sun et al., 2013): Let $\forall x_1 \in \mathbb{R}, \forall x_2 \in \mathbb{R}, \eta \geq 1$ and $o_\eta = 2^{\eta-1}$, the inequality satisfies $|x_1 + x_2|^\eta \leq o_\eta(|x_1|^\eta + |x_2|^\eta)$.

Lemma 4. (Jiang & Praly, 1998): There exists a smooth function $l(\cdot)$ for any $\epsilon > 0$ satisfying $|x| \leq xl(x) + \epsilon$, where $l(0) = 0$.

Lemma 5. (Jiang & Praly, 1998): There exist a nonnegative smooth function $\hat{f}$ with $\frac{\partial \hat{f}}{\partial x(0)}$ for any $\epsilon > 0$ and any continuous function $f(x)$ satisfying $f(0) = 0$, the inequality $|f(x)| \leq \hat{f}(x) + \epsilon$ holds.

Lemma 6. (Qian & Lin, 2001): There is a set Ω_e defined as $\Omega_e = \{e | |e| \leq 0.8814v\}$, where $v > 0$ is a constant. For any $e \notin \Omega_e$, we have $[1 - \tanh^2(\frac{e}{v})] \leq 0$.

Lemma 7. (Sanner & Slotine, 1992): The unknown nonlinear function $f(\Theta) : \mathbb{R}^n \rightarrow \mathbb{R}$ is approximated by RBFNNs, it is described as $f(\Theta) = W^T \beta(\Theta)$, where $\Theta \in \Omega_\Theta \subset \mathbb{R}^o$ is the input vector with o denoting the neural network input dimension and $W = [\varpi_1, \dots, \varpi_s]^T \in \mathbb{R}^s$ means weight vector with $s > 1$ denoting the NNs node number. $\beta(\Theta) = [q_1(\Theta), \dots, q_s(\Theta)]^T \in \mathbb{R}^s$ stands for basis function vector and $q_i(\Theta)$ is chosen as the Gaussian-like function: $q_i(\Theta) = e^{-\frac{(\Theta - v_i)^T(\Theta - v_i)}{\eta^2}}$. In the above relationship, $v_i = [v_{i1}, \dots, v_{io}]^T$ ($1 \leq i \leq s$) is center of the receptive domain, and η is the width of the Gaussian function. Based on Sanner and Slotine (1992), if $f(\Theta)$ is a continuous function and the elements of $f(\Theta)$ pertain to a compact set Ω_Θ , NNs $W^{*T} \beta(\Theta)$ is: $f(\Theta) = W^{*T} \beta(\Theta) + \zeta(\Theta), \forall \Theta \in \Omega_\Theta$, where the $\zeta(\Theta)$ is the approximation error. For any given constant $\zeta^* > 0, \zeta(\Theta) < \zeta^*$. In the above relationship, the optimal weight vector is W^* shown as $W^* = \arg \min_{W \in \mathbb{R}^s} \sup_{\Theta \in \Omega_\Theta} |f(\Theta) - W^{*T} \beta(\Theta)|$.

Lemma 8. If V is an exponentially input-to-state practically predefined-time stable Lyapunov function for the z -system $\dot{z} = q(z, y)$. In other words, (2) holds, for any constant $\frac{\pi}{rT_c} \in (0, \frac{\pi}{rT_c})$, any initial instant $t_0 > 0$, any initial situation $z^0 = z(t_0)$, and $a^0 > 0$, any function $\bar{\gamma}(x_1) \geq \gamma(|x_1|)$, there is a finite $T^0 = T^0(\frac{\pi}{rT_c}, a^0, z^0) \geq 0$, a nonnegative function $D(t_0, t)$ defined for all $t \geq t_0$ and a signal can be described as

$$\dot{a} = -\frac{\pi}{rT_c} a^{1+\frac{r}{2}} - \frac{\pi}{rT_c} a^{1-\frac{r}{2}} + \bar{\gamma}(x_1) + d_0 \quad (3)$$

where $a(t_0) = a^0$, d_0 is a constants defined later, such that $D(t_0, t) = 0$ for all $t \geq t_0 + T^0$ and

$$V(z(t)) \leq a(t) + \frac{1}{2} a^2(t) + D(t_0, t) \quad (4)$$

for all $t \geq t_0$; the solutions are defined. Without loss of generality, assuming $\bar{\gamma}$ is a smooth function which can be described as: $\bar{\gamma}(s) = s^2 \mu_0(s^2)$. Then, the dynamics signal is rewritten as: $\dot{a} = -\frac{\pi}{rT_c} a^{1+\frac{r}{2}} - \frac{\pi}{rT_c} a^{1-\frac{r}{2}} + x_1^2 \mu_0(x_1^2) + d_0$, where $a(t_0) = a^0$ and μ_0 is a nonnegative smooth function.

Proof. The proof process is detailed in Appendix. $\square$

2.3. Predefined-time dynamics surface and controller design

In this part, the predefined-time controller is designed based on the backstepping control framework. To achieve control target, the following error coordinates are defined.

$$e_1 = x_1 - y_d,$$

$$e_i = x_i - w_i,$$

$$z_i = w_i - \tau_{i-1} \quad (5)$$

where $i = 2, \dots, n$. e_1 is the tracking error, e_i denotes the surface error, z_i is the dynamic surface filter output error and τ_{i-1} is the virtual controller. w_i and τ_{i-1} can satisfy the following a first-order filter

$$\begin{aligned} \dot{w}_i = & -\left(\frac{1}{2}\right)^{1-\frac{r}{2}} \frac{\pi}{rT_c} z_i^{1-r} - \left(\frac{1}{2}\right)^{1+\frac{r}{2}} \frac{(3n)^{\frac{r}{2}} \pi}{rT_c} z_i^{1+r} \\ & - \frac{1}{2} z_i - \rho_{i-1,1} z_i^p - \rho_{i-1,2} z_i^{\frac{p+1}{p_1-1}-1} \end{aligned} \quad (6)$$

where $w_i(0) = \tau_{i-1}(0)$ and $0 < r < 1, T_c > 0, \pi > 0$ and $\rho_{i-1,1} > 0$ are design parameters. $\rho_{i-1,2}$ is design function that will be given later ($i = 2, \dots, n$). Subsequently, we outline the update laws $\dot{\hat{\theta}}_s$ ($s = 1, \dots, n$), the virtual control signals τ_1 and τ_j ($j = 2, \dots, n-1$) and actual control signal u , which are given as follows

$$\dot{\hat{\theta}}_s = \frac{b_s}{2k_s^2} e_s^{2(p-p_s+1)} \beta_s^T \beta_s - \frac{b_s \pi}{rT_c} \hat{\theta}_s - \frac{b_s \pi}{rT_c} \hat{\theta}_s^3 \quad (7)$$

$$\begin{aligned} \tau_1 = & \left(-\frac{1}{2k_1^2} e_1^{p-p_1+1} \hat{\theta}_1 \beta_1^T \beta_1 - \frac{\pi}{rT_c} e_1^{p_1+\frac{r}{2}(p+1)} \right. \\ & \left. - \frac{\pi}{rT_c} e_1^{p_1-\frac{r}{2}(p+1)} \right)^{\frac{1}{p_1}} \end{aligned} \quad (8)$$

$$\begin{aligned} \tau_j = & \left(-\frac{1}{2k_j^2} e_j^{p-p_j+1} \hat{\theta}_j \beta_j^T \beta_j - \frac{\pi}{rT_c} e_j^{p_j+\frac{r}{2}(p+1)} \right. \\ & \left. - \frac{\pi}{rT_c} e_j^{p_j-\frac{r}{2}(p+1)} \right)^{\frac{1}{p_j}} \end{aligned} \quad (9)$$

$$\begin{aligned} u = & \frac{1}{\eta_n} \left(\left(-\frac{1}{2k_n^2} e_n^{p-p_n+1} \hat{\theta}_n \beta_n^T \beta_n - \frac{\pi}{rT_c} e_n^{p_n+\frac{r}{2}(p+1)} \right. \right. \\ & \left. \left. - \frac{\pi}{rT_c} e_n^{p_n-\frac{r}{2}(p+1)} \right)^{\frac{1}{p_n}} - \xi(t) \right) \end{aligned} \quad (10)$$

where $k_1, \dots, k_n$ and $b_1, \dots, b_n$ are positive parameters.

Remark 2. The dynamic surface filters in Peng and Dubay (2019) and Hu and Ma (2025) fail to guarantee predefined-time convergence. In contrast, the proposed dynamic surface filter in (6) ensures convergence within a user-specified time, a property that cannot be achieved by the above dynamic surface filters.

2.4. Controller design methodology

Step 1. The Lyapunov function candidate is chosen as $V_1 = \frac{e_1^{p-p_1+2}}{p-p_1+2} + \frac{1}{2b_1} \tilde{\theta}_1^2 + \frac{a}{\gamma_0}$, where $\tilde{\theta}_1 = \theta_1^* - \hat{\theta}_1$, γ_0 is a positive design constant. Next, by considering Assumption 3, then $\dot{V}_1$ is obtained as

$$\begin{aligned} \dot{V}_1 = & e_1^{p-p_1+1} (x_2^{p_1} + f_1(x_1) - \dot{y}_d) - \frac{1}{b_1} \tilde{\theta}_1 \dot{\hat{\theta}}_1 \\ & + \frac{1}{\gamma_0} \left(-\frac{\pi}{rT_c} a^{1+\frac{r}{2}} - \frac{\pi}{rT_c} a^{1-\frac{r}{2}} + x_1^2 \mu_0(x_1^2) \right. \\ & \left. + d_0 \right) + |e_1^{p-p_1+1}| (\psi_{11}(|y|) + \psi_{12}(|z|)) \end{aligned} \quad (11)$$

From Assumption 4, we know that γ_1 the k_∞ -function, then γ_1^{-1} is also the k_∞ -function. Then, the inequality (4) reveals that

$$\psi_{12}(|z|) \leq \psi_{12}[\gamma_1^{-1}(a(t) + D(t_0, t))] \quad (12)$$

By defining $\psi_{12} \circ \gamma_1^{-1}(a(t) + D(t_0, t)) = \psi_{12}[\gamma_1^{-1}(a(t) + D(t_0, t))]$, yields: $\psi_{12}(|z|) \leq \psi_{12} \circ \gamma_1^{-1}(a(t) + D(t_0, t))$. For any t , we get:

$\min\{2a(t), 2D(t_0, t)\} \leq a(t) + D(t_0, t) \leq \max\{2a(t), 2D(t_0, t)\}$. Next, we deal with the last term of inequality (11) by Young's inequality, then

$$\begin{aligned} & |e_1^{p-p_1+1}|(\psi_{11}(|y|) + \psi_{12}(|z|)) \\ & \leq |e_1^{p-p_1+1}|\psi_{11}(|y|) + |e_1^{p-p_1+1}|\psi_{12} \circ \gamma_1^{-1}(2a) \\ & \quad + |e_1^{p-p_1+1}|\psi_{12} \circ \gamma_1^{-1}(2D(t_0, t)) \\ & \leq |e_1^{p-p_1+1}|\psi_{11}(|y|) + |e_1^{p-p_1+1}|\psi_{12} \circ \gamma_1^{-1}(2a) \\ & \quad + \frac{1}{4}e_1^{2(p-p_1+1)} + (\psi_{12} \circ \gamma_1^{-1}(2D(t_0, t)))^2 \end{aligned} \quad (13)$$

Applying Lemma 4 and Lemma 5, we can get

$$\begin{aligned} & |e_1^{p-p_1+1}|\psi_{11}(|y|) + |e_1^{p-p_1+1}|\psi_{12} \circ \gamma_1^{-1}(2a) \\ & \leq e_1^{p-p_1+1}\hat{\psi}_{11}(y) + e_1^{p-p_1+1}\hat{\psi}_{12}(a)g_{11}(y, a) \\ & \quad + \varepsilon_{12} + \varepsilon_{11} + e_1^{p-p_1+1}g_{12}(y) + \varepsilon_{12} \end{aligned} \quad (14)$$

where $\hat{\psi}_{11}$ and $\hat{\psi}_{12}$ are two suitable smooth functions satisfying $\hat{\psi}_{11}(0) = 0$ and $\hat{\psi}_{12}(0) = 0$. Further, we can rewrite inequality (13) as follows by defining $d_1(t_0, t) = (\psi_{12} \circ \gamma_1^{-1}(2D(t_0, t)))^2$ and $\hat{\psi}_{13}(y, a) = \hat{\psi}_{12}(a)g_{11}(y, a) + g_{12}(y)$.

$$\begin{aligned} & |e_1^{p-p_1+1}|(\psi_{11}(|y|) + \psi_{12}(|z|)) \leq e_1^{p-p_1+1}\hat{\psi}_{11}(y) + \varepsilon_{11} \\ & \quad + 2\varepsilon_{12} + d_1(t_0, t) + \frac{e_1^{2(p-p_1+1)}}{4} + e_1^{p-p_1+1}\hat{\psi}_{13}(y, a) \end{aligned} \quad (15)$$

Substituting the above inequality into Eq. (11) yields

$$\begin{aligned} \dot{V}_1 & \leq e_1^{p-p_1+1}(x_2^{p_1} - \tau_1^{p_1} + \tau_1^{p_1} + f_1(x_1) - \dot{y}_d \\ & \quad + \hat{\psi}_{11}(y) + \hat{\psi}_{13}(y, a) + \frac{1}{4}e_1^{p-p_1+1}) - \frac{1}{b_1}\bar{\theta}_1\dot{\hat{\theta}}_1 \\ & \quad + \varepsilon_{11} + 2\varepsilon_{12} + d_1(t_0, t) + \frac{1}{\gamma_0}\left(-\frac{\pi}{\hat{r}\hat{T}_c}a^{1+\frac{r}{2}}\right. \\ & \quad \left.- \frac{\pi}{\hat{r}\hat{T}_c}a^{1-\frac{r}{2}} + d_0 + x_1^2\mu_0(x_1^2)\right) \end{aligned} \quad (16)$$

With the support of Lemma 2, it yields

$$\begin{aligned} e_1^{p-p_1+1}(x_2^{p_1} - \tau_1^{p_1}) & \leq h_1 e_1^{p+1} + \rho_{11} e_2^{p+1} \\ & \quad + \rho_{11} z_2^{p+1} + \rho_{12} e_2^{\frac{p+1}{p_1}} + \rho_{12} z_2^{\frac{p+1}{p_1}} \end{aligned} \quad (17)$$

where $h_1 > 0$ is a parameter, $\rho_{11} = 2^{p+1} \frac{p_1^2}{p+1} \left(\frac{h_1(p+1)}{p_1(p-p_1+1)} \times \frac{1}{2^{p_1}} \right)^{-\frac{p-p_1+1}{p_1}}$ and $\rho_{12} = 2^{\frac{p+1}{p_1}} p_1(2^{p_1-1} + 1) \left(\frac{h_1(p+1)}{p_1(p-p_1+1)} \times \frac{1}{(2^{p_1-1}+2)} \right)^{-\frac{p-p_1+1}{p_1}} \frac{p_1}{p+1}$. Meanwhile, we define $\bar{f}_1 = f_1 - \dot{y}_d + \hat{\psi}_{11}(y) + \hat{\psi}_{13}(y, a) + h_1 e_1^{p+1} + \frac{2x_1^2\mu_0(x_1^2)}{\gamma_0 e_1^{p-p_1+1}} \tanh^2\left(\frac{e_1^{p-p_1+1}}{v}\right) - \frac{1}{2}e_1^{p-p_1+1}$. Then, based on NNs, we can get the following relationship: $e_1^{p-p_1+1}\bar{f}_1 \leq \frac{\theta_1^* \beta_1^T \beta_1}{2k_1^2} e_1^{2(p-p_1+1)} + \frac{k_1^2}{2} + \frac{e_1^{2(p-p_1+1)}}{4} + \zeta_1^{*2}$, where $\theta_1^* = \|W_1^*\|^2$. ζ_1 can satisfy $|\zeta_1| \leq \zeta_1^*$ and ζ_1^* is a positive parameter. Thus, $\dot{V}_1$ becomes

$$\begin{aligned} \dot{V}_1 & \leq e_1^{p-p_1+1}(\tau_1^{p_1} + \frac{e_1^{p-p_1+1}}{2k_1^2}\bar{\theta}_1\beta_1^T\beta_1) + \frac{\bar{\theta}_1}{b_1}\left(\frac{b_1}{2k_1^2}\right. \\ & \quad \times e_1^{2(p-p_1+1)}\beta_1^T\beta_1 - \bar{\theta}_1) + \varepsilon_{11} + 2\varepsilon_{12} + d_1(t_0, t) \\ & \quad - \frac{\pi}{\gamma_0\hat{r}\hat{T}_c}a^{1+\frac{r}{2}} - \frac{\pi}{\gamma_0\hat{r}\hat{T}_c}a^{1-\frac{r}{2}} + \frac{d_0}{\gamma_0} + \frac{x_1^2\mu_0(x_1^2)}{\gamma_0} \\ & \quad \times (1 - 2 \tanh^2(\frac{e_1^{p-p_1+1}}{v})) + \rho_{11}e_2^{p+1} + \rho_{11}z_2^{p+1} \\ & \quad + \rho_{12}e_2^{\frac{p+1}{p_1}} + \rho_{12}z_2^{\frac{p+1}{p_1}} + \zeta_1^{*2} + \frac{k_1^2}{2} \end{aligned} \quad (18)$$

Considering virtual control τ_1 from (8) and adaptive law $\dot{\hat{\theta}}_1$ from (7), one has

$$\dot{V}_1 \leq -\frac{\pi}{rT_c}e_1^{(1+\frac{r}{2})(p+1)} - \frac{\pi}{rT_c}e_1^{(1-\frac{r}{2})(p+1)} + \frac{\pi}{rT_c}\bar{\theta}_1\hat{\theta}_1$$

$$\begin{aligned} & + \frac{\pi}{rT_c}\bar{\theta}_1\hat{\theta}_1^3 + \rho_{11}e_2^{p+1} + \rho_{12}e_2^{\frac{p+1}{p_1}} + \rho_{12}z_2^{\frac{p+1}{p_1}} \\ & - \frac{\pi}{\gamma_0\hat{r}\hat{T}_c}a^{1+\frac{r}{2}} - \frac{\pi}{\gamma_0\hat{r}\hat{T}_c}a^{1-\frac{r}{2}} + \frac{d_0}{\gamma_0} + \frac{x_1^2\mu_0(x_1^2)}{\gamma_0} \\ & \times (1 - 2 \tanh^2(\frac{e_1^{p-p_1+1}}{v})) + C_1 + \rho_{11}z_2^{p+1} \end{aligned} \quad (19)$$

where $C_1 = \varepsilon_{11} + 2\varepsilon_{12} + d_1(t_0, t) + \zeta_1^{*2} + \frac{k_1^2}{2}$.

Step i ($2 \leq i \leq n-1$). A Lyapunov function is chosen: $V_i = V_{i-1} + \frac{e_i^{p-p_i+2}}{p-p_i+2} + \frac{1}{2b_i}\bar{\theta}_i^2 + \frac{1}{2}z_i^2$, where $\bar{\theta}_i = \theta_i^* - \hat{\theta}_i$. $\dot{V}_i$ is calculated as

$$\begin{aligned} \dot{V}_i & = \dot{V}_{i-1} + e_i^{p-p_i+1}(x_{i+1}^{p_i} + f_i + \triangle_i + \tau_i^{p_i} \\ & \quad - \tau_i^{p_i} - \dot{w}_i) - \frac{\bar{\theta}_i}{b_i}\dot{\hat{\theta}}_i + z_i(\dot{w}_i - \dot{\tau}_{i-1}) \end{aligned} \quad (20)$$

Similar to the Eqs. (12)-(15) in the Step 1, one has

$$\begin{aligned} e_i^{p-p_i+1}\triangle_i & \leq e_i^{p-p_i+1}\hat{\psi}_{i1}(y) + \varepsilon_{i1} + 2\varepsilon_{i2} \\ & \quad + d_i(t_0, t) + \frac{1}{4}e_i^{2(p-p_i+1)} + e_i^{p-p_i+1}\hat{\psi}_{i3}(y, a) \end{aligned} \quad (21)$$

where $\hat{\psi}_{i1}$ and $\hat{\psi}_{i2}$ are two suitable smooth functions such that $\hat{\psi}_{i1}(0)=0$ and $\hat{\psi}_{i2}(0)=0$. $d_i(t_0, t) = (\psi_{i2} \circ \gamma_i^{-1}(2D(t_0, t)))^2$ and $\hat{\psi}_{i3}(y, a) = \hat{\psi}_{i2}(a)g_{i1}(y, a) + g_{i2}(y)$. Similarly, define $\bar{f}_i =$

$f_i + \hat{\psi}_{i1} + \hat{\psi}_{i3} + h_i e_i^{p_i} - \frac{1}{2}e_i^{p-p_i+1} - \dot{w}_i + \rho_{i-1,1}e_i^{p_i} + \rho_{i-1,2}e_i^{\frac{p+1}{p_{i-1}}-(p-p_i+1)}$. Analogous to the inequality scaling process in Step 1, by using Young's inequality, NNs and Lemma 2, one can derive the following result

$$\begin{aligned} e_i^{p-p_i+1}(x_{i+1}^{p_i} - \tau_i^{p_i}) & \leq h_i e_i^{p+1} + \rho_{i1} e_{i+1}^{p+1} + \rho_{i1} z_{i+1}^{p+1} \\ & \quad + \rho_{i2} e_{i+1}^{\frac{p+1}{p_i}} + \rho_{i2} z_{i+1}^{\frac{p+1}{p_i}} \end{aligned} \quad (22)$$

$$e_i^{p-p_i+1}\bar{f}_i \leq \frac{\theta_i^* \beta_i^T \beta_i}{2k_i^2} e_i^{2(p-p_i+1)} + \frac{k_i^2}{2} + \frac{e_i^{2(p-p_i+1)}}{4} + \zeta_i^{*2} \quad (23)$$

$$-z_i \dot{\tau}_{i-1} \leq \frac{1}{2}z_i^2 + \frac{1}{2}\dot{\tau}_{i-1}^2 \quad (24)$$

where $h_i > 0$ is a parameter and $\theta_i^* = \|W_i^*\|^2$. ζ_i can satisfy $|\zeta_i| \leq \zeta_i^*$ and ζ_i^* is a positive parameter. $\rho_{i1} = 2^{p+p_i} \frac{p_i^2}{p+1} \left(\frac{h_i(p+1)}{p_i(p-p_i+1)} \times \frac{1}{2^{p_i}} \right)^{-\frac{p-p_i+1}{p_i}}$ and

$\rho_{i2} = \left(\frac{h_i(p+1)}{p_i(p-p_i+1)} \times \frac{1}{(2^{p_i-1}+2)} \right)^{-\frac{p-p_i+1}{p_i}} \frac{p_i}{p+1} 2^{\frac{p+1}{p_i}} (2^{p_i-1} + 1)$. From the above inequalities (22)-(24) and with the support of virtual control signal τ_i in (9), adaptive law $\dot{\hat{\theta}}_i$ in (7) and the dynamics surface filter signal $\dot{w}_i$ (6), $\dot{V}_i$ can be written as

$$\begin{aligned} \dot{V}_i & \leq -\sum_{j=1}^i \frac{\pi}{rT_c} e_j^{(1+\frac{r}{2})(p+1)} - \sum_{j=1}^i \frac{\pi}{rT_c} e_j^{(1-\frac{r}{2})(p+1)} - \sum_{j=2}^i \frac{\pi}{rT_c} \\ & \quad \times \left(\frac{z_j^2}{2}\right)^{1-\frac{r}{2}} + \sum_{j=1}^i \frac{\pi}{rT_c} \bar{\theta}_j \dot{\theta}_j + \sum_{j=1}^i \frac{\pi}{rT_c} \bar{\theta}_j \dot{\theta}_j^3 + \sum_{j=1}^i C_j \\ & \quad - \frac{\pi}{\gamma_0\hat{r}\hat{T}_c}a^{1+\frac{r}{2}} - \frac{\pi}{\gamma_0\hat{r}\hat{T}_c}a^{1-\frac{r}{2}} + \frac{d_0}{\gamma_0} + \rho_{i2}e_{i+1}^{\frac{p+1}{p_i}} + \rho_{i2}z_{i+1}^{\frac{p+1}{p_i}} \\ & \quad + \frac{x_1^2\mu_0(x_1^2)}{\gamma_0}(1 - 2 \tanh^2(\frac{e_1^{p-p_1+1}}{v})) + \rho_{i1}z_{i+1}^{p+1} \\ & \quad + \sum_{j=2}^i \frac{\tau_{j-1}^2}{2} + \rho_{i1}e_{i+1}^{p+1} - \sum_{j=2}^i \frac{(3n)\pi}{rT_c} \left(\frac{1}{2}z_j^2\right)^{1+\frac{r}{2}} \end{aligned} \quad (25)$$

where $C_i = \varepsilon_{i1} + 2\varepsilon_{i2} + d_i(t_0, t) + \zeta_i^{*2} + \frac{k_i^2}{2}$.

Step n . Design the Lyapunov function as: $V_n = V_{n-1} + \frac{e_n^{p-p_n+2}}{p-p_n+2} + \frac{1}{2b_n}\bar{\theta}_n^2 + \frac{1}{2}z_n^2$, where $\bar{\theta}_n = \theta_n^* - \hat{\theta}_n$. Following this analysis, we derive the time derivative

$$\dot{V}_n = \dot{V}_{n-1} + e_n^{p-p_n+1}((\eta_n u + \xi)^{p_n} + f_n + \triangle_n$$

$$- \dot{w}_n) - \frac{\tilde{\theta}_n}{b_n} \dot{\hat{\theta}}_n + z_n (\dot{w}_n - \dot{\tau}_{n-1}) \quad (26)$$

Repeat the operation similar to inequality (12)-(15) in the Step 1 we have

$$\begin{aligned} e_n^{p-p_n+1} \triangleleft_n &\leq e_n^{p-p_n+1} \hat{\psi}_{n1}(y) + \varepsilon_{n1} + 2\varepsilon_{n2} \\ &+ d_n(t_0, t) + \frac{1}{4} e_n^{2(p-p_n+1)} + e_n^{p-p_n+1} \hat{\psi}_{n3}(y, a) \end{aligned} \quad (27)$$

where $\hat{\psi}_{n1}$ and $\hat{\psi}_{n2}$ are two suitable smooth functions and they make $\hat{\psi}_{n1}(0) = 0$ and $\hat{\psi}_{n2}(0) = 0$ hold. $d_n(t_0, t) = (\psi_{n2} \circ \gamma_n^{-1}(2D(t_0, t)))^2$ and $\hat{\psi}_{n3}(y, a) = \hat{\psi}_{n2}(a)g_{n1}(y, a) + g_{n2}(y)$. The NNs are selected to approximate $\tilde{f}_n$, where $\tilde{f}_n = f_n + \hat{\psi}_{n1} + \hat{\psi}_{n3} - \dot{w}_n - \frac{1}{2} e_n^{p-p_n+1} + \rho_{n-1,1} e_n^{p_n} + \rho_{n-1,2} e_n^{\frac{p+1}{2} - (p-p_n+1)}$. Then, we get: $e_n^{p-p_n+1} f_n \leq \frac{\theta_n^* \hat{\theta}_n^T \beta_n}{2k_n^2} e_n^{2(p-p_n+1)} + \frac{k_n^2}{2} + \frac{e_n^{2(p-p_n+1)}}{4} + \zeta_n^{*2}$, where $\theta_n^* = \|W_n^*\|^2$. ζ_n can satisfy $|\zeta_n| \leq \zeta_n^*$ and ζ_n^* is a positive parameter. Then, we deal with the term $-z_n \dot{\tau}_{n-1}$ from (26) by Young's inequalities, one has: $-z_n \dot{\tau}_{n-1} \leq \frac{1}{2} z_n^2 + \frac{1}{2} \dot{\tau}_{n-1}^2$. Finally, combining the above inequality, the actual controller u in (10), the adaptive law $\dot{\hat{\theta}}_n$ in (7) and the dynamics surface filter signal $\dot{w}_n$ from (6), it yields

$$\begin{aligned} \dot{V}_n &\leq - \sum_{j=1}^n \frac{\pi}{rT_c} e_j^{(1+\frac{r}{2})(p+1)} - \sum_{j=1}^n \frac{\pi}{rT_c} e_j^{(1-\frac{r}{2})(p+1)} + \sum_{j=1}^n C_j \\ &+ \sum_{j=1}^n \frac{\pi}{rT_c} \tilde{\theta}_j \hat{\theta}_j + \sum_{j=1}^n \frac{\pi}{rT_c} \tilde{\theta}_j \hat{\theta}_j^3 - \sum_{j=2}^n \frac{\pi}{rT_c} \left(\frac{1}{2} z_j^2\right)^{1-\frac{r}{2}} \\ &- \frac{\pi a^{1+\frac{r}{2}}}{\gamma_0 \hat{r} T_c} - \frac{\pi a^{1-\frac{r}{2}}}{\gamma_0 \hat{r} T_c} - \sum_{j=2}^n \frac{(3n)^{\frac{r}{2}} \pi}{rT_c} \left(\frac{1}{2} z_j^2\right)^{1+\frac{r}{2}} \\ &+ \frac{x_1^2 \mu_0(x_1^2)}{\gamma_0} (1 - 2 \tanh^2(\frac{e_1^{p-p_1+1}}{v})) + \frac{d_0}{\gamma_0} \end{aligned} \quad (28)$$

where $C_n = \varepsilon_{n1} + 2\varepsilon_{n2} + d_n(t_0, t) + \zeta_n^{*2} + \frac{k_n^2}{2} + \sum_{j=1}^{n-1} \frac{1}{2} |\tau_j^*|$. τ_j^* is a constant, which can satisfy $\tau_j \leq |\tau_j^*|$ from Li et al. (2019b).

2.5. Stability analysis

Theorem 1. The controlled HNSs (1) satisfied Assumptions 1–3, the actual controller (10), the virtual controllers (8)-(9), the update laws (7) can be designed. The following properties hold: all signals of the HNSs (1) are predefined-time bounded and the tracking error can converge to the small neighborhood near the origin.

Proof. of Theorem 1: Based on Lemma 2, Young's inequality, and $\tilde{\theta} = \theta^* - \hat{\theta}$, one has

$$\begin{aligned} \frac{\pi}{rT_c} \tilde{\theta}_j \hat{\theta}_j &\leq -\frac{\pi \tilde{\theta}_j^2}{2rT_c} + \frac{\pi (\theta_j^*)^2}{2rT_c} \\ &\leq -\frac{\pi}{rT_c} \left(\frac{\tilde{\theta}_j^2}{2b_j}\right)^{1-\frac{r}{2}} + \frac{\pi (\theta_j^*)^2}{2rT_c} + \frac{\pi}{2T_c} \left(\frac{2b_j}{2-r}\right)^{\frac{-2-r}{r}} \end{aligned} \quad (29)$$

□

According to the complete cubic formula, we obtain

$$\frac{\pi \tilde{\theta}_j^3}{rT_c} \leq \frac{\pi}{rT_c} \tilde{\theta}_j (\theta_j^*)^3 - \frac{3\pi}{rT_c} \tilde{\theta}_j^2 (\theta_j^*)^2 - \frac{3\pi}{rT_c} \tilde{\theta}_j^3 \theta_j^* - \frac{\pi \tilde{\theta}_j^4}{rT_c} \quad (30)$$

Subsequently, applying Lemma 2, we obtain

$$\frac{\pi}{rT_c} \tilde{\theta}_j (\theta_j^*)^3 \leq \frac{3\pi}{rT_c} \tilde{\theta}_j^2 (\theta_j^*)^2 + \frac{\pi}{12rT_c} (\theta_j^*)^4 \quad (31)$$

$$\frac{3\pi}{rT_c} \tilde{\theta}_j^3 \theta_j^* \leq \frac{\pi}{2rT_c} \tilde{\theta}_j^4 + \frac{3\pi}{4rT_c} \left(\frac{2}{9}\right)^{-3} (\theta_j^*)^4 \quad (32)$$

Combining formulas (31), (32), formula (30) can be rearranged as

$$\frac{\pi}{rT_c} \tilde{\theta}_j \hat{\theta}_j^3 \leq -\frac{\pi \tilde{\theta}_j^4}{2rT_c} + \frac{\pi (\theta_j^*)^4}{12rT_c} + \frac{3\pi}{4rT_c} \left(\frac{2}{9}\right)^{-3} (\theta_j^*)^4 \quad (33)$$

Subsequently, we apply Lemma 2 to deal with the term $-\frac{\pi \tilde{\theta}_j^4}{2rT_c}$ in formula (33), obtaining: $-\frac{\pi \tilde{\theta}_j^4}{2rT_c} \leq -\frac{(3n)^{\frac{r}{2}} \pi}{rT_c} \left(\frac{\tilde{\theta}_j^2}{2b_j}\right)^{1+\frac{r}{2}} + c_{\theta_{j1}}$, where $c_{\theta_{j1}} = \frac{(3n)^{\frac{r}{2}} \pi}{2rT_c} (1 - \frac{r}{2}) \left(\frac{4b_j^2}{(1+\frac{r}{2})(3n)^{\frac{r}{2}}}\right)^{-\frac{2+r}{2-r}}$. Based on the aforementioned inequality transformation, we rewrite formula (33) as

$$\frac{\pi \tilde{\theta}_j \hat{\theta}_j^3}{rT_c} \leq -\frac{(3n)^{\frac{r}{2}} \pi}{rT_c} \left(\frac{\tilde{\theta}_j^2}{2b_j}\right)^{1+\frac{r}{2}} + c_{\theta_j} \quad (34)$$

where $c_{\theta_j} = c_{\theta_{j1}} + \frac{\pi (\theta_j^*)^4}{12rT_c} + \frac{3\pi}{4rT_c} \left(\frac{2}{9}\right)^{-3} (\theta_j^*)^4$. Next, we deal with the terms $\sum_{j=1}^n \frac{\pi}{rT_c} e_j^{(1+\frac{r}{2})(p+1)}$ and $\sum_{j=1}^n \frac{\pi}{rT_c} e_j^{(1-\frac{r}{2})(p+1)}$ in formula (28). We still use the conclusion of Lemma 2, and then we obtain

$$-\frac{\pi}{rT_c} e_j^{(1+\frac{r}{2})(p+1)} \leq -\frac{(3n)^{\frac{r}{2}} \pi}{rT_c} e_j^{(1+\frac{r}{2})(p-p_j+2)} + E_{j1} \quad (35)$$

$$-\frac{\pi}{rT_c} e_j^{(1-\frac{r}{2})(p+1)} \leq -\frac{\pi}{rT_c} e_j^{(1-\frac{r}{2})(p-p_j+2)} + E_{j2} \quad (36)$$

where $E_{j1} = \frac{(p_j-1)\pi}{(p+1)rT_c} \left(\frac{1}{p-p_j+2}\right)^{1+\frac{r}{2}} ((p-p_j+2)^{\frac{r}{2}} (p+1) \times ((3n)^{\frac{r}{2}}))^{\frac{-p-p_j+2}{p_j-1}}$

and $E_{j2} = \frac{(p_j-1)\pi}{(p+1)rT_c} \left(\frac{1}{p-p_j+2}\right)^{1-\frac{r}{2}} ((p-p_j+2)^{-\frac{r}{2}} (p+1))^{\frac{-p-p_j+2}{p_j-1}}$. In the above (29)-(36), $j = 1, 2, \dots, n$ is the index. By combining the above inequalities (29), (34), (35), (36), we have

$$\begin{aligned} \dot{V}_n &\leq - \sum_{j=1}^n \frac{(3n)^{\frac{r}{2}} \pi}{rT_c} \left(\frac{e_j^{p-p_j+2}}{p-p_j+2}\right)^{1+\frac{r}{2}} - \sum_{j=2}^n \frac{(3n)^{\frac{r}{2}} \pi}{rT_c} \\ &\times \left(\frac{1}{2} z_j^2\right)^{1+\frac{r}{2}} - \sum_{j=1}^n \frac{\pi}{rT_c} \left(\frac{e_j^{p-p_j+2}}{p-p_j+2}\right)^{1-\frac{r}{2}} - \frac{\pi a^{1-\frac{r}{2}}}{rT_c} \\ &+ \frac{x_1^2 \mu_0(x_1^2)}{\gamma_0} (1 - 2 \tanh^2(\frac{e_1^{p-p_1+1}}{v})) + C_s \\ &- \sum_{j=1}^n \frac{(3n)^{\frac{r}{2}} \pi}{rT_c} \left(\frac{\tilde{\theta}_j^2}{2b_j}\right)^{1+\frac{r}{2}} - \sum_{j=1}^n \frac{\pi}{rT_c} \left(\frac{\tilde{\theta}_j^2}{2b_j}\right)^{1-\frac{r}{2}} \\ &- \frac{(3n)^{\frac{r}{2}} \pi}{rT_c} a^{1+\frac{r}{2}} - \sum_{j=2}^n \frac{\pi}{rT_c} \left(\frac{1}{2} z_j^2\right)^{1-\frac{r}{2}} \end{aligned} \quad (37)$$

where $\gamma_0 \hat{r} T_c = \frac{rT_c}{(3n)^{\frac{r}{2}}}$, $C_s = \frac{d_0}{\gamma_0} + \sum_{j=1}^n C_j + \sum_{j=1}^n E_{j1} + \sum_{j=1}^n E_{j2} + \frac{2-r}{2} \left(\frac{1}{8+4r}\right)^{-\frac{2+r}{2-r}} \frac{n\pi}{rT_c} + \sum_{j=1}^n c_{\theta_j} + \sum_{j=1}^n \frac{\pi}{2rT_c} \theta_j^{*2} + \sum_{j=1}^n \frac{\pi}{rT_c} \left(\frac{2b_j}{2-r}\right)^{\frac{-2-r}{r}}$. Finally, according to Lemmas 3–4 in Ma et al. (2025), we can obtain the following expression

$$\begin{aligned} \dot{V}_n &\leq - \sum_{j=1}^n \sum_{s=2}^n \left[\frac{(3n)^{\frac{r}{2}} \pi}{rT_c} \left(\frac{e_j^{p-p_j+2}}{p-p_j+2}\right)^{1+\frac{r}{2}} + a^{1+\frac{r}{2}} \right. \\ &+ \left. \left(\frac{\tilde{\theta}_j^2}{2b_j}\right)^{1+\frac{r}{2}} + \left(\frac{1}{2} z_s^2\right)^{1+\frac{r}{2}} \right] - \sum_{j=1}^n \sum_{s=2}^n \left[\frac{\pi}{rT_c} \left(a^{1-\frac{r}{2}} \right. \right. \\ &+ \left. \left. \left(\frac{e_j^{p-p_j+2}}{p-p_j+2}\right)^{1-\frac{r}{2}} + \left(\frac{\tilde{\theta}_j^2}{2b_j}\right)^{1-\frac{r}{2}} + \left(\frac{1}{2} z_s^2\right)^{1-\frac{r}{2}} \right) \right] \\ &+ \frac{x_1^2 \mu_0(x_1^2)}{\gamma_0} (1 - 2 \tanh^2(\frac{e_1^{p-p_1+1}}{v})) + C_s \\ &\leq -\frac{\pi}{rT_c} V_n^{1-\frac{r}{2}} - \frac{\pi}{rT_c} V_n^{1+\frac{r}{2}} + C_s \\ &+ \frac{x_1^2 \mu_0(x_1^2)}{\gamma_0} (1 - 2 \tanh^2(\frac{e_1^{p-p_1+1}}{v})) \end{aligned} \quad (38)$$

Next, we proceed to analyze the boundedness of $\frac{x_1^2 \mu_0(x_1^2)}{\gamma_0} (1 - 2 \tanh^2(\frac{e_1^{p-p_1+1}}{v}))$ by considering the following two scenarios. First,

we analyze the first case. Suppose that $e_1^{p-p_1+1} \in \Omega_{e_1}$, where $\Omega_{e_1} = \{e_1 | |e_1^{p-p_1+1}| < 0.8814v\}$ and $v > 0$ is a constant. Based on the boundedness of e_1 and y_d , the boundedness of x_1 can be obtained. It then follows directly that the boundedness of $\frac{x_1^2 \mu_0(x_1^2)}{\gamma_0} (1 - 2 \tanh^2(\frac{e_1^{p-p_1+1}}{v}))$ is guaranteed. Suppose that c^* denotes its bound, then the following inequality holds: $\dot{V}_n \leq -\frac{\pi}{rT_c} V_n^{1-\frac{r}{2}} - \frac{\pi}{rT_c} V_n^{1+\frac{r}{2}} + d$, where $d = C_s + c^*$. Subsequently, we proceed to analyze the second situation. Suppose that $e_1^{p-p_1+1} \notin \Omega_{e_1}$. With the support of Lemma 6, it follows that $\frac{x_1^2 \mu_0(x_1^2)}{\gamma_0} (1 - 2 \tanh^2(\frac{e_1^{p-p_1+1}}{v})) \leq 0$. Therefore, we obtain: $\dot{V}_n \leq -\frac{\pi}{rT_c} V_n^{1-\frac{r}{2}} - \frac{\pi}{rT_c} V_n^{1+\frac{r}{2}} + C_s$. Overall, the inequality (39) holds with $C_\sigma = \max\{d, C_s\}$.

$$\dot{V}_n \leq -\frac{\pi}{rT_c} V_n^{1-\frac{r}{2}} - \frac{\pi}{rT_c} V_n^{1+\frac{r}{2}} + C_\sigma \quad (39)$$

With the help of the boundedness of V_n , the boundedness of $a, e_1, \dots, e_n, \tilde{\theta}_1, \dots, \tilde{\theta}_n$ and $z_2, \dots, z_n$ can be got. On the basis of the boundedness of e_1 and y_d , the state x_1 is bounded. According to $\tilde{\theta}_1 = \theta_1^* - \hat{\theta}_1$, we see that $\hat{\theta}_1$ is bounded due to θ_1^* and $\tilde{\theta}_1$ are bounded. The Eq. (8) guarantees that virtual control law τ_1 remains bounded, as this property is inherited from the bounded nature of both $\hat{\theta}_1$ and e_1 . Subsequently, the boundedness of w_2 can be obtained from coordinate transformation. Then, based on $x_2 = e_2 + w_2$, we know the boundedness of x_2 . Finally, based on similar analyses mentioned above, we receive that $x_3, \dots, x_n, \hat{\theta}_2, \dots, \hat{\theta}_n$ and u are bounded.

The guidelines for parameter selection are as follows. The parameter h_s in the predefined-time dynamic surface filter (6) is set to $h_s > 0$ for $s = 1, 2, \dots, n$. Parameters $b_s > 0$ are chosen to determine the adaptive laws $\hat{\theta}_s$ in (7). The parameters $\hat{r}$ and $\hat{T}_c$ in the predefined-time dynamic signal (3) are selected such that $0 < \frac{\pi}{\hat{r}\hat{T}_c} < \frac{\pi}{rT_c}$. Appropriate parameters $0 < r < 1, k_s > 0$, and $T_c > 0$ are chosen to construct the virtual and actual controllers in (8), (9), and (10). Notably, within an appropriate range, the smaller the parameter T_c is selected, the shorter the time it takes for the system to reach a steady state. However, it should be noted that the parameter T_c affects the magnitude of the control input. Specifically, selecting a smaller T_c generally leads to faster convergence, but at the cost of increased control effort and potentially higher actuator burden. In contrast, a larger T_c results in smaller control input magnitude, while sacrificing convergence speed. Therefore, in practical applications, T_c should be carefully tuned to balance convergence performance and actuator constraints.

Remark 3. From Lemma 1, the settling time upper bound depends only on the design parameter T_c and is independent of the initial conditions. Different initial conditions or scenarios do not change the settling time upper bound. Therefore, based on the stability analysis, the settling time upper bound of the proposed dynamic signal a is determined solely by T_c .

Remark 4. Predefined-time control is of significance in real-time human-computer interaction systems. With the widespread application of NNs in real-time interaction fields such as brain-computer interfaces (Chen et al., 2025), the system requirements for response timeliness and stability have been continuously increasing. Therefore, control methods with the characteristic of predefined convergence time possess important engineering application value.

3. Illustrative examples

Example 1. Consider the following nonlinear system

$$\begin{aligned} \dot{z} &= -z + 0.3x_1^2 + 0.25, \\ \dot{x}_1 &= x_2^{p_1} + f_1(x_1) + \Delta_1, \\ \dot{x}_2 &= u_f^{p_2} + f_2(x_1, x_2) + \Delta_2 \end{aligned}$$

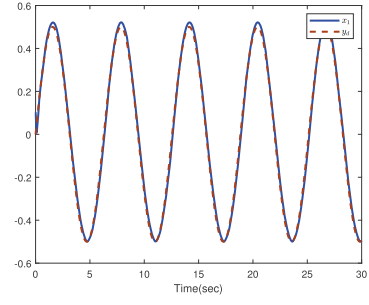


Fig. 1. The tracking curves (Example 1).

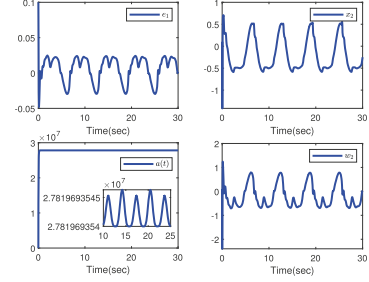


Fig. 2. The tracking error e_1 , system state x_2 , the dynamics signal a and dynamics surface signal w_2 (Example 1).

Table 1

Table I: Performance Comparison of Different Control Methods.

Evaluation Items	Our method	Existing method
Convergence time	$\sqrt{2}$ s	4.93 s
Root mean square of tracking error	5.039×10^{-3}	6.073×10^{-2}
Peak value of the control input	4.5960×10^3	3.7365×10^3

where $f_1(x_1) = x_1^2 + 0.2 \ln(1 + x_1^2)$, $f_2(x_1, x_2) = 0.5x_1x_2 + 0.1(e^{0.5x_1} - 1) + 0.1 \ln(1 + x_2^2)$, $\Delta_1 = zx_1 \sin(x_1)$, $\Delta_2 = zx_1x_2$, $p_1 = 1$, $p_2 = 3$ and $p = \max\{p_1, p_2\} = 3$. The reference signal is $y_d = 0.5 \sin(t)$. To meet Assumption 3, we select the following conditions: $\psi_{i1}(s) = s^2$ and $\varphi_{i2}(s) = s^4$ ($i = 1, 2$). Assumption 3 holds for the z -subsystem with $V(z) = z^2$, $\frac{\pi}{\hat{r}\hat{T}_c} = 4\pi$, $\mu_0(x_1^2) = 2.5x_1^2$, $\tilde{r}(x_1) = 2.5x_1^4$, $d_0 = 7.9369 \times 10^8$, $\gamma_1(z) = 0.5z^2$ and $\gamma_2(z) = 1.5z^2$. The actuator fault is given as: $u_f = 0.1u + 0.1 \sin(0.1t)$. The initial values are displayed: $x_1(0) = 0.1$, $a(0) = 1$, $z(0) = 0.3$, $x_2(0) = w_2(0) = 0.1$, $\hat{\theta}_1(0) = \hat{\theta}_2(0) = 0.5$. The design parameters for both the controller and adaptive laws can be determined as follows: $r = 0.06$, $T_c = 4$, $h_1 = 8$, $b_1 = b_2 = 0.1$, $k_1 = k_2 = 0.1$, $\gamma_0 = 0.91$.

The results of the simulation are provided in Figs. 1–3. From Fig. 1, it is showed the trajectories of the system state and given desired signal. Fig. 2 shows the tracking error e_1 , systems state x_2 , the dynamics signal a and dynamics surface signal w_2 . It can be obtained that they are bounded within predefined time. It is obvious from Figs. 3 show that the actual controll signal u , unmodeled dynamic signal z and adaptive parameters $\hat{\theta}_1$ and $\hat{\theta}_2$ are predefined-time bounded. The above simulation results show that the designed controller achieves tracking of the desired trajectory. The system states, dynamic surface filter signals and control inputs all converge rapidly and oscillate stably, demonstrating good dynamic performance. The adaptive parameters converge quickly to the origin, thus verifying the effectiveness of the proposed control method.

Example 2. The single-link robotic manipulator system under consideration in Example 2 originates from Sun et al. (2016). The corresponding model can be written as: $\partial \epsilon \ddot{\lambda}(t) = u(t) - \partial \mu \sin(\lambda(t)) - \kappa \epsilon \dot{\lambda}(t)$, where $\lambda, \dot{\lambda}, \ddot{\lambda}$ are link position, velocity, and acceleration, respectively. ∂, ϵ denote mass and the length of the link, respectively. μ is gravity acceleration

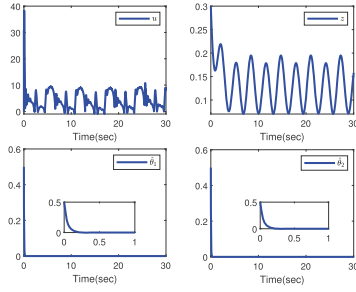


Fig. 3. The actual control signal u , the unmodeled dynamics signal z , the adaptive parameters $\hat{\theta}_1$ and $\hat{\theta}_2$ (Example 1).

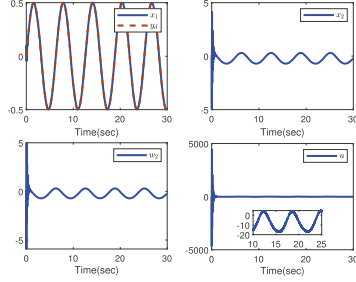


Fig. 4. The tracking curves, systems states x_2 and dynamics surface signal w_2 actual control signal u (Example 2).

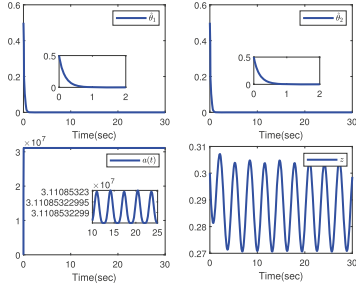


Fig. 5. The adaptive parameters $\hat{\theta}_1, \hat{\theta}_2$, the dynamic signal a and the unmodeled dynamics signal z (Example 2).

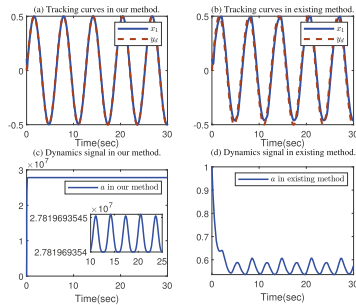


Fig. 6. The existing method and our method.

constant. κ denotes unknown viscous friction coefficient. The parameter values are shown as $\theta = 1$ kg, $\varepsilon = 2$ m, $\mu = 9.8$ m/s² and $\kappa = 1$. Define $x_1 = \lambda$ and $x_2 = \dot{\lambda}$, we consider the following mathematical model

$$\begin{aligned} \dot{x}_1 &= x_2 + \kappa_1, \\ \dot{x}_2 &= \frac{1}{\theta\varepsilon}u - \frac{\mu}{\varepsilon}\sin(x_1) - \frac{\kappa}{\theta}x_2 + \kappa_2 \end{aligned} \quad (40)$$

where $\kappa_1 = 0.2\cos(x_1^2)$ and $\kappa_2 = 0.2\cos(x_1x_2^2)$ are external disturbances to validate the disturbance rejection capability of the proposed method. The design parameters are set as: $T_c = 1$, $\gamma_0 = 0.81$ and $\hat{\tau}_c = 0.07$, and the

parameter settings satisfy the requirements of Lemma 8. From Lemma 1, the settling time is $\sqrt{2}s$. In addition, the other designed parameters, the reference signal, the actuator fault, the initial values and the unmodeled dynamics are selected to be identical to those in Example 1. Fig. 4 shows the tracking curves, system state x_2 , dynamics surface signal w_2 and actual control signal u . From Fig. 5, the simulation results of adaptive parameters $\hat{\theta}_1, \hat{\theta}_2$, dynamics signal a and unmodeled dynamics signal z can be obtained. As shown in the simulation results, the controller enables the system output to quickly track the target signal within predefined time, indicating strong tracking performance and good robustness even with unmodeled dynamics and external disturbances. Both state variables and dynamics surface filter signals converge rapidly to stable values, demonstrating stable system response and effective filter design. The actual control input remains bounded throughout, and the adaptive parameters converge quickly, verifying the effectiveness of the adaptive algorithm. The curves of dynamic signals and unmodeled dynamics further confirm the controller's ability to effectively suppress these factors.

Using the system of Example 2 as an illustration, we applied the method outlined in Ai et al. (2024), selecting the same initial state, reference signal, actuator failure, and unmodeled dynamics as those in Example 2 without external disturbances. The resulting simulation outcomes are depicted in Fig. 6. Obviously, in Fig. 6, we can see that the convergence time of the predefined-time dynamic signal control method proposed in this paper is significantly faster than existing fixed-time dynamic signal control strategies in Ai et al. (2024). In addition, Table 1 is provided to more clearly illustrate the results of the comparative experiments. Among them, the settling time of the existing method (Ai et al., 2024) is obtained by selecting parameters $\gamma_1 = 0.998$, $\iota = \frac{1}{3}$, which can also be visually observed from the simulation results in Fig. 6. Combining Fig. 6 and Table 1, it can be seen that the proposed method significantly outperforms existing methods in both convergence speed and tracking accuracy, with superior dynamic response. Although this performance improvement entails a slight increase in control input and dynamic signal amplitude, it is a reasonable engineering trade-off, as the substantial gains in dynamic response and tracking precision fully justify this minor rise in control effort, reflecting a fundamental balance inherent in control engineering. Its practical significance lies in reminding designers to balance performance with constraints such as actuator limits, energy consumption, and robustness. The phenomenon observed in this study is generally acceptable in practice, as it provides clear performance gains.

4. Conclusion

The adaptive neural predefined-time control approach has been presented for HNSs with dynamics uncertainties and actuator fault. The unknown nonlinear functions of the HNSs have been approximated via using NNs. With the support of the proposed novel dynamics surface technology, the “complexity explosion problem” has been avoided. Further, a novel predefined-time dynamics signal has been proposed that can deal with the dynamics uncertainties of the controlled systems. According to predefined-time theory and backstepping method, it has been demonstrated that all signals of considered systems can be bounded within predefined time interval. The simulation case has been implied the validity of the control method. Despite improved convergence and tracking, balancing speed and control gain in predefined-time control remains a future research direction

CRediT authorship contribution statement

Jiawei Ma: Writing – review & editing, Writing – original draft, Software, Methodology, Investigation, Formal analysis, Conceptualization; **Huagang Zhang:** Visualization, Supervision, Resources, Funding acquisition; **Juan Zhang:** Visualization, Supervision, Investigation; **Le Wang:** Visualization, Supervision, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Appendices

Proof. In the proof process, we will discuss it in different situations.

(1) Firstly, we will discuss the first scenario, which is $0 < a < 1$ and $0 < V < 1$. From [Assumption 4](#), we get

$$\dot{V} \leq -\frac{\pi}{rT_c} V^{1+\frac{r}{2}}(z) - \frac{\pi}{rT_c} V^{1-\frac{r}{2}}(z) + \gamma(|x_1|) + \bar{d}_0 \quad (\text{A.1})$$

□

With the help of [Lemma 2](#), we get $\frac{\pi}{rT_c} a^{1-\frac{r}{2}} \leq \frac{\pi}{2\hat{r}T_c} a^2 + c_1$, where $c_1 = (\frac{1}{2} + \frac{r}{4})(\frac{2}{2-r})^{\frac{r-2}{r+2}} \frac{\pi}{rT_c}$. By applying the relationship between $\gamma(|x_1|)$ and $\bar{r}(x_1)$, the equality (3) and defining $d_0 = \bar{d}_0 + \frac{\pi}{rT_c} P_1 + c_1$, one has

$$\dot{V} \leq -\frac{2\pi}{rT_c} V + \dot{a} + \frac{2\pi}{rT_c} a + \frac{\pi a^2}{2\hat{r}T_c} \quad (\text{A.2})$$

On the basis of Gronwall's Lemma, one has

$$(e^{\frac{2\pi}{rT_c} t} V)' \leq e^{\frac{2\pi}{rT_c} t} \dot{a} + e^{\frac{2\pi}{rT_c} t} \frac{2\pi}{rT_c} a \quad (\text{A.3})$$

Integrating (A.1) over $[t, t_0]$, then

$$\begin{aligned} \int_{t_0}^t (e^{\frac{2\pi}{rT_c} t} V)' dt &\leq \int_{t_0}^t e^{\frac{2\pi}{rT_c} t} \dot{a}(t) dt + \int_{t_0}^t e^{\frac{2\pi}{rT_c} t} \frac{2\pi}{rT_c} a(t) dt \\ &\quad + \int_{t_0}^t e^{\frac{2\pi}{rT_c} t} \frac{\pi a(t)^2}{2\hat{r}T_c} dt \end{aligned} \quad (\text{A.4})$$

Multiplying both sides of the inequality (A.2) by $e^{-\frac{2\pi}{rT_c} t}$ yields

$$\begin{aligned} V(z(t)) - e^{-\frac{2\pi}{rT_c} (t-t_0)} V(z(t_0)) &\leq \int_{t_0}^t \dot{a}(t) dt \\ &\quad + \int_{t_0}^t \frac{2\pi}{rT_c} a(t) dt + \int_{t_0}^t \frac{\pi}{2\hat{r}T_c} a^2(t) dt \end{aligned} \quad (\text{A.5})$$

Next, multiplying (A.3) by $e^{\frac{\pi}{rT_c} t}$, then

$$\begin{aligned} e^{\frac{\pi}{rT_c} t} V(z(t)) - e^{\frac{\pi}{rT_c} t - \frac{2\pi}{rT_c} (t-t_0)} V(z(t_0)) &\leq a(t) e^{\frac{\pi}{rT_c} t} \\ &\quad - a(t_0) e^{\frac{\pi}{rT_c} t_0} + \frac{1}{2} a^2(t) e^{\frac{\pi}{rT_c} t} - \frac{1}{2} a^2(t_0) e^{\frac{\pi}{rT_c} t_0} \end{aligned} \quad (\text{A.6})$$

Further, multiplying (A.4) by $e^{-\frac{\pi}{rT_c} t}$, one has

$$\begin{aligned} V(z(t)) &\leq a(t) + \frac{1}{2} a^2(t) - a(t_0) e^{-\frac{\pi}{rT_c} (t-t_0)} \\ &\quad - \frac{1}{2} a^2(t_0) e^{\frac{\pi}{rT_c} (t-t_0)} + e^{-\frac{2\pi}{rT_c} (t-t_0)} V(z(t_0)) \end{aligned} \quad (\text{A.7})$$

Define $D(t_0, t) = \max\{0, e^{-\frac{2\pi}{rT_c} (t-t_0)} V(z(t_0)) - \frac{1}{2} a^2(t_0) e^{\frac{\pi}{rT_c} (t-t_0)} - a(t_0) e^{-\frac{\pi}{rT_c} (t-t_0)}\}$, the relationship holds: $V(z(t)) \leq a(t) + \frac{1}{2} a^2(t) + D(t_0, t)$.

(2) Next, we will discuss the remaining scenarios.

When $0 < a < 1$ and $V > 1$, we define $d_0 = \bar{d}_0 + c_1 + c_2$, where $c_2 = \frac{r}{2}(\frac{1}{2-r})^{\frac{r-2}{r}} \frac{\pi}{rT_c}$. When $0 < a < 1$ and $V = 1$, d_0 can be defined as $d_0 = \bar{d}_0 + c_1$. When $a > 1$ and $0 < V < 1$, we give the definition of d_0 is $d_0 = \bar{d}_0 + \frac{\pi}{rT_c} P_1 + c_3$, where $c_3 = (\frac{1}{2} - \frac{r}{4})(\frac{2}{2+r})^{\frac{-2-r}{2-r}} \frac{\pi}{rT_c}$. When $a > 1$ and $V > 1$, we define $d_0 = \bar{d}_0 + c_2 + c_3$. When $a > 1$ and $V = 1$, d_0 is $d_0 = \bar{d}_0 + c_3$. When $a = 1$ and $0 < V < 1$, d_0 can be defined as $d_0 = \bar{d}_0 + \frac{\pi}{rT_c} P_1$. When $a = 1$ and $V > 1$, the definition of d_0 is $d_0 = \bar{d}_0 + c_2$. When $a = 1$ and $V = 1$, we define $d_0 = \bar{d}_0$.

Similar to formulas (A.1)-(A.7), when defining $D(t_0, t) = \max\{0, e^{-\frac{2\pi}{rT_c} (t-t_0)} V(z(t_0)) - \frac{1}{2} a^2(t_0) e^{\frac{\pi}{rT_c} (t-t_0)} - a(t_0) e^{-\frac{\pi}{rT_c} (t-t_0)}, e^{-\frac{\pi}{rT_c} (t-t_0)} \times V(z(t_0)) - \frac{1}{2} a^2(t_0) e^{\frac{\pi}{rT_c} (t-t_0)} - a(t_0) e^{-\frac{\pi}{rT_c} (t-t_0)}\}$, both the above situations can make $V(z(t)) \leq a(t) + \frac{1}{2} a^2(t) + D(t_0, t)$ hold true.

In conclusion, we take $P_2 = \max\{\frac{\pi}{rT_c} P_1, c_2\}$, $P_3 = \max\{c_1, c_3\}$, and define $d_0 = \bar{d}_0 + P_2 + P_3$, $D(t_0, t) = \max\{0, e^{-\frac{2\pi}{rT_c} (t-t_0)} V(z(t_0)) - \frac{1}{2} a^2(t_0) e^{\frac{\pi}{rT_c} (t-t_0)} - a(t_0) \times e^{-\frac{\pi}{rT_c} (t-t_0)}, e^{-\frac{\pi}{rT_c} (t-t_0)} V(z(t_0)) - \frac{1}{2} a^2(t_0) e^{\frac{\pi}{rT_c} (t-t_0)} - a(t_0) \times e^{-\frac{\pi}{rT_c} (t-t_0)}\}$. At this point, $V(z(t)) \leq a(t) + \frac{1}{2} a^2(t) + D(t_0, t)$ holds.

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